

antenna and user end allow all radio transmit, receive, signal generation, modulation, demodulation, timing, control, coding and decoding functions to be performed in software.

A generic block diagram is shown in Fig. 2. Although no current architecture can claim to meet the full potential of a software radio, many successful software radio architectures have been designed that have met their design goals.

Ideally, SDR products must be flexible towards operational standards and independent from carrier frequencies. Advances in areas such as DSP and digital converter performance, field-programmable gate array (FPGA) density and object-oriented programming have given a base for more generic hardware and highly capable and flexible software to perform most of the radio processing tasks.

SDR is a very useful technology as it allows one radio platform to service multiple radio standards. The availability, cost and the diverse market have,

however, limited its deployment.

Five stages in evolution of SDR

SDR Forum—an international, non-profit organisation promoting the development of SDR—defines software-defined radio as “a collection of hardware and software technologies that enable reconfigurable systems architecture for wireless networks and user terminals.”

As the possible development trend and the best advancing path, the SDR Forum defines five stages of development. Stage 0 is traditional radio implementation in hardware. Stage 1 implements the control features for multiple hardware elements in software. Stage 2, SDR, implements modulation and baseband processing in software but allows for multiple-frequency, fixed-function RF hardware. Stage 3, ideal software radio (ISR), extends programmability through the radio frequency (RF) with analogue conversion at the antenna. Stage 4, ultimate software

radio (USR), provides for fast (millisecond) transitions between communication protocols in addition to digital processing capability.

Thus these five stages of radio development are based on a software platform, the foundation of which is the traditional radio. Eventually, the aim is full digitisation so as to make it an ideal software radio.

Key sections of SDR

A software radio consists of smart antenna, high-speed data conversion and high-speed signal processing sections.

Smart antenna and RF stage. The antenna and front end of RF are the hardware input and output ends of software radio, so these cannot be replaced with software.

A smart antenna is an antenna array system aided by some smart algorithm designed to adapt to different signal environments. Smart antennae mitigate fading through diversity reception and beam forming while minimising interference through spatial filtering.

The software radio spans multiple bands, up to multiple octaves per band with uniform shape and low losses to provide access to available service bands. Thus, wideband multi-band smart antenna technology is essential.

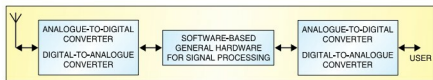


Fig. 1: Block diagram of an ideal SDR

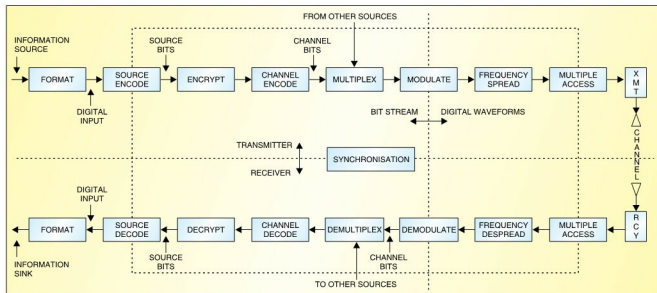


Fig. 2: Block diagram of generic SDR